



MNS University of Agriculture, Multan

Institute of Computing

PROJECT



CS-501 Artificial Intelligence

Title:

Breast Cancer Prediction (ML)

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1. Introduction

Breast cancer is one of the most common cancers worldwide. Early diagnosis is critical for increasing survival rates. Traditional diagnosis involves a pathologist visually examining biopsy slides, which can be time-consuming and subject to human error. This project aims to automate the classification of tumors as either Malignant (cancerous) or Benign (non-cancerous) using Machine Learning techniques.

2. Goal

The goal of this project is to develop a machine learning model that can analyze clinical features of a breast mass (such as radius, texture, and smoothness) and accurately predict the diagnosis. This tool is intended to assist medical professionals in making faster and more reliable decisions.

3. Methodology

We approached the problem using a standard Supervised Learning workflow:

3.1 Dataset

- **Source:** Breast Cancer Wisconsin Dataset (obtained from Kaggle).
- **Features:** The dataset includes 30+ features computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. Key features include Radius, Texture, Perimeter, Area, and Smoothness.
- **Target Variable:** The diagnosis column contains two classes: 'M' (Malignant) and 'B' (Benign).

3.2 Data Preprocessing

Before training, the data was cleaned and prepared using Python libraries:

- **Cleaning:** Irrelevant columns such as id and empty columns (Unnamed: 32) were removed to prevent errors.
- **Encoding:** The target labels were converted into numerical format: Malignant = 1, Benign = 0.
- **Scaling:** We applied StandardScaler to normalize the data. This ensures that features with large values (like "Area") do not dominate features with small values (like "Smoothness").
- We utilized **Logistic Regression**, a powerful classification algorithm suitable for binary outcomes. The dataset was split into a **Training Set (80%)** for the model to learn and a **Testing Set (20%)** to evaluate its performance.

3.3 Model Selection

- We utilized Logistic Regression, a powerful classification algorithm suitable for binary outcomes. The dataset was split into a Training Set (80%) for the model to learn and a Testing Set (20%) to evaluate its performance.

4. Tools & Technologies

- **Programming Language:** Python
- **Data Analysis:** pandas
- **Machine Learning:** scikit-learn
- **Visualization:** matplotlib, seaborn
- **Version Control:** GitHub

5. Results and Analysis

5.1 Model Accuracy

- The Logistic Regression model achieved an accuracy of approximately 95% - 98% (*Replace this with your exact score from the code*) on the testing dataset. This indicates a high level of reliability in distinguishing between malignant and benign tumors.

5.2 Confusion Matrix

The confusion matrix below illustrates the model's performance: (*Insert the confusion_matrix.png image here*)

- **True Positives:** Correctly identified Malignant cases.
- **True Negatives:** Correctly identified Benign cases.
- **False Negatives:** Dangerous cases where the model missed cancer (we aimed to minimize these).

5.3 Feature Correlation

- We analyzed the relationships between different clinical features: (*Insert the correlation_heatmap.png image here*)
- The heatmap shows that features like Radius and Perimeter are highly correlated, which is expected as larger tumors tend to have larger perimeters.

6. Conclusion

The project successfully demonstrates that Machine Learning can be effectively applied to medical diagnosis. The Logistic Regression model performed with high accuracy, suggesting that such tools can serve as a valid second opinion for doctors.

7. Project Repository

The complete source code and documentation for this project are available on GitHub:

<https://github.com/malikuzairasghar/Breast-Cancer-Prediction>